

Quadratic Functions (2.1)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $(2, 0)$

_____ State the vertex of $f(x) = (x - 3)^2 + 2$.

ANSWER: $\{-3, 3\}$

_____ Find the x -value that minimizes $f(x) = x^2 - 8x + 1$.

ANSWER: $x = 3$

_____ State the y -intercept of $f(x) = x^2 - 4x + 7$.

ANSWER: $\{-2, -1, 1, 2\}$

_____ Does the parabola $f(x) = -2x^2 + 3x$ open up or down?

ANSWER: (3, 2) # _____ State the axis of symmetry of $f(x) = x^2 - 6x + 5$.	ANSWER: 0 # _____ Solve $x^4 - 5x^2 + 4 = 0$.
ANSWER: (-1, -1) # _____ Solve $x^2 - 9 = 0$.	ANSWER: down # _____ Write the vertex of $y = x^2 - 4x + 4$ in vertex form's terms.
ANSWER: (0, 7) # _____ Find the vertex of $f(x) = x^2 + 2x$.	ANSWER: 4 # _____ How many x-intercepts does $f(x) = x^2 + 4$ have?